

COSMOS COMPUTER

jQuery

jQuery is a small, light-weight and fast JavaScript library. It is cross-platform and supports different types of browsers. It is also referred as "write less do more" because it takes a lot of common tasks that requires many lines of JavaScript code to accomplish, and binds them into methods that can be called with a single line of code whenever needed. It is also very useful to simplify a lot of the complicated things from JavaScript, like AJAX calls and DOM manipulation.

- **jQuery is a small, fast and lightweight JavaScript library.**
- **jQuery is platform-independent.**
- **jQuery means "write less do more".**
- **jQuery simplifies AJAX call and DOM manipulation.**

Link

```
<script  
src="https://ajax.googleapis.com/ajax/libs/jquery/3.5.  
1/jquery.min.js">  
</script>
```

EXAMPLE OF EFFECTS

No	Topic	Program
1	Demo	<pre> <!DOCTYPE html> <html> <head> <script src="https://ajax.googleapis.com/ajax/libs/jquery/3.5.1/jquery.mi n.js"> </script> <script> \$(document).ready(function(){ \$("p").click(function(){ \$(this).hide(); }); }); </script> </head> <body> <p>If you click on me, I will disappear.</p> <p>Click me away!</p> <p>Click me too!</p> </body> </html> </pre>
2	Button	<pre> <!DOCTYPE html> <html> <head> <script src="https://ajax.googleapis.com/ajax/libs/jquery/3.5.1/jquery.mi n.js"> </script> <script> \$(document).ready(function(){ \$("button").click(function(){ \$("p").hide(); }); }); </script> </pre>

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		<pre> </head> <body> <h2>This is a heading</h2> <p>This is a paragraph.</p> <p>This is another paragraph.</p> <button>Click me</button> </body> </html> </pre>
3	Hide() & Show()	<pre> <!DOCTYPE html> <html> <head> <script src="https://ajax.googleapis.com/ajax/libs/jquery/3.5.1/jquery.mi n.js"></script> <script> \$(document).ready(function(){ \$("#hide").click(function(){ \$("p").hide(); }); \$("#show").click(function(){ \$("p").show(); }); }); </script> </head> <body> <p>If you click on the "Hide" button, I will disappear.</p> <button id="hide">Hide</button> <button id="show">Show</button> </body> </html> </pre>

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4.	Fade()	<pre><!DOCTYPE html> <html> <head> <script src="https://ajax.googleapis.com/ajax/libs/jquery/3.5.1/jquery.mi n.js"></script> <script> \$(document).ready(function(){ \$("button").click(function(){ \$("#div1").fadeIn(); \$("#div2").fadeIn("slow"); \$("#div3").fadeIn(3000); }); }); </script> </head> <body> <p>Demonstrate fadeIn() with different parameters.</p> <button>Click to fade in boxes</button>

 <div id="div1" style="width:80px;height:80px;display:none;background- color:red;"></div>
 <div id="div2" style="width:80px;height:80px;display:none;background- color:green;"></div>
</pre>
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		<pre> <div id="div3" style="width:80px;height:80px;display:none;background- color:blue;"></div> </body> </html> </pre>
5.	Slide()	Syntax: <code>\$(selector).slideDown(speed,callback);</code> <pre> <!DOCTYPE html> <html> <head> <script src="https://ajax.googleapis.com/ajax/libs/jquery/3.5.1/jquery.mi n.js"></script> <script> \$(document).ready(function(){ \$("#flip").click(function(){ \$("#panel").slideDown("slow"); }); }); </script> <style> #panel, #flip { padding: 5px; text-align: center; background-color: #e5eccc; border: solid 1px #c3c3c3; } #panel { padding: 50px; display: none; } </style> </head> <body> <div id="flip">Click to slide down panel</div> <div id="panel">Hello world!</div> </body> </pre>

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		</html>
6.	animate()	The jQuery animate() method is used to create custom animations. Syntax: <code>\$(selector).animate({params},speed,callback);</code> <!DOCTYPE html> <html> <head> <script src="https://ajax.googleapis.com/ajax/libs/jquery/3.5.1/jquery.min.js"></script> <script> \$(document).ready(function(){ \$("button").click(function(){ \$("div").animate({left: '250px'}); }); }); </script> </head> <body> <button>Start Animation</button> <div style="background:red;height:100px;width:100px;position:absolute;"></div> </body> </html>
7.	Stop()	The jQuery stop() method is used to stop an animation or effect before it is finished. The stop() method works for all jQuery effect functions, including sliding, fading and custom animations. Syntax: <code>\$(selector).stop(stopAll,goToEnd);</code>

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```
<!DOCTYPE html>

<html>

<head>

<script
src="https://ajax.googleapis.com/ajax/libs/jquery/3.5.1/jquery.mi
n.js"></script>

<script>

$(document).ready(function(){

    $("#flip").click(function(){

        $("#panel").slideDown(5000);

    });

    $("#stop").click(function(){

        $("#panel").stop();

    });

});

</script>
<style>
#panel, #flip {
    padding: 5px;
    font-size: 18px;
    text-align: center;
    background-color: pink;
    color: white;
    border: solid 1px #666;
    border-radius: 3px;
}

#panel {
    padding: 50px;
    display: none;
}
</style>
</head>
```

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		<pre> <body>

 <button id="stop">Stop sliding</button> <div id="flip">Click to slide down panel</div> <div id="panel">Hello world!</div> </body> </html> </pre>
8.	Callback	• JavaScript statements are executed line by line. However, with effects, the next line of code can be run even though the effect is not finished. This can create errors. • To prevent this, you can create a callback function. • A callback function is executed after the current effect is finished. • syntax: <code>\$(selector).hide(speed,callback);</code> <pre> <!DOCTYPE html> <html> <head> <script src="https://ajax.googleapis.com/ajax/libs/jquery/3.5.1/jquery.mi n.js"></script> <script> \$(document).ready(function(){ \$("button").click(function(){ \$("p").hide("slow", function(){ alert("The paragraph is now hidden"); }); }); }); </pre>

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		<pre></script> </head> <body> <button>Hide</button> <p>This is a paragraph with little content.</p> </body> </html></pre>
--	--	---

EXAMPLE OF HTML

No	Topic
1.	Get(): • One very important part of jQuery is the possibility to manipulate the DOM. • jQuery comes with a bunch of DOM related methods that make it easy to access and manipulate elements and attributes. • DOM = Document Object Model • Three simple, but useful, jQuery methods for DOM manipulation are: • text() - Sets or returns the text content of selected elements • html() - Sets or returns the content of selected elements (including HTML markup) • val() - Sets or returns the value of form fields <u>1.text() and html():</u> <pre> <!DOCTYPE html> <html> <head> <script src="https://ajax.googleapis.com/ajax/libs/jquery/3.5.1/jquery.min.js"></ script> <script> \$(document).ready(function(){ \$("#btn1").click(function(){ alert("Text: " + \$("#test").text()); }); \$("#btn2").click(function(){ alert("HTML: " + \$("#test").html()); }); }); </script> </head> <body> <p id="test">This is some bold text in a paragraph.</p> <button id="btn1">Show Text</button> </pre>

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```
<button id="btn2">Show HTML</button>
```

```
</body>
```

```
</html>
```

2.val():

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<script
```

```
src="https://ajax.googleapis.com/ajax/libs/jquery/3.5.1/jquery.min.js"></script>
```

```
<script>
```

```
$(document).ready(function(){
```

```
    $("button").click(function(){
```

```
        alert("Value: " + $("#test").val());
```

```
    });
```

```
});
```

```
</script>
```

```
</head>
```

```
<body>
```

```
<p>Name: <input type="text" id="test" value=""></p>
```

```
<button>Show Value</button>
```

```
</body>
```

```
</html>
```

2. Set():

```
<html>
```

```
<head>
```

```
<script
```

```
src="https://ajax.googleapis.com/ajax/libs/jquery/3.5.1/jquery.min.js"></script>
```

```
>
```

```
<script>
```

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```
$(document).ready(function(){
    $("#btn1").click(function(){
        $("#test1").text("Hello world!");
    });
    $("#btn2").click(function(){
        $("#test2").html("<b>Hello world!</b>");
    });
    $("#btn3").click(function(){
        $("#test3").val("Dolly Duck");
    });
});
</script>
</head>
<body>

<p id="test1">This is a paragraph1.</p>
<p id="test2">This is paragraph2.</p>

<p>Input field: <input type="text" id="test3" value=""></p>

<button id="btn1">Set Text</button>
<button id="btn2">Set HTML</button>
<button id="btn3">Set Value</button>

</body>
</html>
```

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3.

Add():

```
<!DOCTYPE html>
<html>
<head>
<script
src="https://ajax.googleapis.com/ajax/libs/jquery/3.5.1/jquery.min.js"></script
>
<script>
$(document).ready(function(){
    $("#btn1").click(function(){
        $("p").append(" <b>Appended text</b>.");
    });

    $("#btn2").click(function(){
        $("ol").append("<li> item added</li>");
    });
});
</script>
</head>
<body>

<p>This is a paragraph.</p>
<p>This is another paragraph.</p>

<ol>
    <li>List item 1</li>
    <li>List item 2</li>
    <li>List item 3</li>
</ol>
```

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```
<button id="btn1">Append text</button>
<button id="btn2">Append list items</button>

</body>
</html>
```

4. Remove():

- **remove()** - Removes the selected element (and its child elements)
- **empty()** - Removes the child elements from the selected element

1.remove():

```
<!DOCTYPE html>
<html>
<head>
<script
src="https://ajax.googleapis.com/ajax/libs/jquery/3.5.1/jquery.min.js"></script>
<script>
$(document).ready(function(){
  $("button").click(function(){
    $("#div1").remove();
  });
});
</script>
</head>
<body>

<div id="div1" style="height:100px;width:300px;border:1px solid
black;background-color:pink;">

This is some text in the div.
<p>cosmos computer.</p>
<p>Bhavnagar.</p>

</div>
<br>

<button>Remove div element</button>
```

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```
</body>
```

```
</html>
```

2.empty():

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<script
```

```
src="https://ajax.googleapis.com/ajax/libs/jquery/3.5.1/jquery.min.js"></script>
```

```
<script>
```

```
$(document).ready(function(){
```

```
  $("button").click(function(){
```

```
    $("#div1").empty();
```

```
  });
```

```
});
```

```
</script>
```

```
</head>
```

```
<body>
```

```
<div id="div1" style="height:100px;width:300px;border:1px solid black;background-color:pink;">
```

```
<p>Cosmos Computer.</p>
```

```
<p>Bhavnagar.</p>
```

```
</div>
```

```
<br>
```

```
<button>Empty the div element</button>
```

```
</body>
```

```
</html>
```

5. **toggleClass()**

- toggleClass() - Toggles between adding/removing classes from the selected elements

```
<!DOCTYPE html>
<html>
<head>
<script
src="https://ajax.googleapis.com/ajax/libs/jquery/3.5.1/jquery.min.js"></script
>
<script>
$(document).ready(function(){
    $("button").click(function(){
        $("p").toggleClass("main");
    });
});
</script>
<style>
.main {
    font-size: 120%;
    color: red;
}
</style>
</head>
<body>

<button>Toggle class "main" for p elements</button>

<p>This is a paragraph.</p>
<p>This is another paragraph.</p>
```


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<p>Note: Click the button more than once to see the toggle effect.</p>

</body>

</html>

<!DOCTYPE html>

<html>

<head>

<script
src="https://ajax.googleapis.com/ajax/libs/jquery/3.5.1/jquery.min.js"></script
>

<script>

\$(document).ready(function(){
 \$("#btn1").click(function(){
 \$("#box").animate({height: "300px"});

});

\$("#btn2").click(function(){
 \$("#box").animate({height: "100px"});

});

});

</script>

</head>

<body>

<button id="btn1">Animate height</button>

<button id="btn2">Reset height</button>

<div id="box"
style="background:red;height:200px;width:100px;margin:6px;"></div>

</body>

</html>

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